

Ex. 26. Find the smallest number that must be added to 1780 to make it a perfect square.

Sol.

4	1780 (42
	16
82	180
	164
	16

$\therefore$ Number to be added = $(43)^2 - 1780 = 1849 - 1780 = 69$.

Ex. 27. If $\sqrt{2} = 1.4142$, find the value of $\frac{\sqrt{2}}{(2+\sqrt{2})}$.

Sol.
$$\frac{\sqrt{2}}{(2+\sqrt{2})} = \frac{\sqrt{2}}{(2+\sqrt{2})} \times \frac{(2-\sqrt{2})}{(2-\sqrt{2})} = \frac{2\sqrt{2}-2}{(4-2)} = \frac{2(\sqrt{2}-1)}{2} = (\sqrt{2}-1) = (1.4142-1)$$

$$= 0.4142.$$

Ex. 28. If $x = \left(\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} \right)$ and $y = \left(\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}} \right)$, find the value of $(x^2 + y^2)$.

Sol.
$$x = \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}-\sqrt{3})} \times \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}+\sqrt{3})} = \frac{(\sqrt{5}+\sqrt{3})^2}{(5-3)} = \frac{5+3+2\sqrt{15}}{2} = 4+\sqrt{15}.$$

$$y = \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}+\sqrt{3})} \times \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})} = \frac{(\sqrt{5}-\sqrt{3})^2}{(5-3)} = \frac{5+3-2\sqrt{15}}{2} = 4-\sqrt{15}.$$

$\therefore x^2 + y^2 = (4+\sqrt{15})^2 + (4-\sqrt{15})^2 = 2[(4)^2 + (\sqrt{15})^2] = 2 \times 31 = 62.$

Ex. 29. Find the value of: $\sqrt{6+\sqrt{6+\sqrt{6+\dots}}}$

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Sol. Let $\sqrt{6+\sqrt{6+\sqrt{6+\dots}}} = x.$

Then, $\sqrt{6+x} = x \Leftrightarrow 6+x = x^2 \Rightarrow x^2 - x - 6 = 0 \Leftrightarrow x^2 - 3x + 2x - 6 = 0$

$\Leftrightarrow x(x-3) + 2(x-3) = 0 \Leftrightarrow (x-3)(x+2) = 0 \Leftrightarrow x = 3.$

$[\because x \neq -2]$

Hence, $\sqrt{6+\sqrt{6+\sqrt{6+\dots}}} = 3.$

Ex. 30. Find the cube root of 2744.

Sol. Method : Resolve the given number as the product of prime factors and take the product of prime factors, choosing one out of three of the same prime factors.

Resolving 2744 as the product of prime factors, we get :

$$2744 = 2^3 \times 7^3.$$

$\therefore \sqrt[3]{2744} = 2 \times 7 = 14.$

2	2744
2	1372
2	686
7	343
7	49
	7

Ex. 31. By what least number 4320 be multiplied to obtain a number which is a perfect cube?

Sol. Clearly, $4320 = 2^3 \times 3^3 \times 2^2 \times 5.$

To make it a perfect cube, it must be multiplied by 2×5^2 i.e., 50.